belinus®

Energywall · H1
v01 · 2026-04
MODEL · HWREW-H1-001

RESIDENTIAL ENERGY STORAGE · HYBRID MODULAR · EDLC · NIB · LITHIUM-FREE

Energywall H1

Unlock the power of nature.

#ResidentialEnergyStorage #HybridModular #NIB #EDLC #LithiumFree

01 **Hybrid modular architecture.** NIB sodium-ion energy + EDLC supercapacitor buffer on the BMS-1 slot standard.

02 **Fully lithium-free.** No lithium, cobalt or nickel — no thermal runaway, inherently safe by design.

03 **10.5 kWh usable per cabinet.** EW-N1-7SC configuration — 7 NIB modules plus a supercapacitor buffer.

04 **> 95 % round-trip efficiency.** 100 % depth of discharge for full use of the storage capacity.

05 **Supercapacitor power layer.** > 50 kW instantaneous peak, > 500,000 cycles.

06 **Scales to 5 cabinets.** Themis EMS auto-discovers modules via NFC — no manual configuration.

20 yr	20 yr	20 yr
PRODUCT WARRANTY	PERFORMANCE WARRANTY	SERVICE WARRANTY

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TECHNICAL SPECIFICATIONS

Energywall H1 · Hybrid Modular Storage

USABLE ENERGY	NOMINAL VOLTAGE	OPERATING TEMP	ROUND-TRIP EFF.
10.5 kWh	46.8 V	−40 → +55 °C	>95 %

GENERAL

Weight	~160 kg (loaded)
Dimensions (W × D × H)	500 × 150 × 1,800 mm
Modules per cabinet	9 (BMS-1 slots)
Cooling	Natural, powder-coated aluminium
Ingress protection	IP54
Operating temperature	−40 °C to +55 °C
Humidity	Non-condensing

PERFORMANCE

Usable DC energy	10.5 kWh (EW-N1-7SC)
Voltage range	36.4 – 53.95 Vdc
Nominal voltage	46.8 Vdc
Internal resistance	< 1 mΩ per cell (1 kHz)
Round-trip efficiency	> 95 %
Depth of discharge	100 %

ARCHITECTURE

Configuration	EW-N1-7SC (7× NIB + supercap)
Module standard	BMS-1 (165 × 470 × 142 mm)
Energy management	Themis EMS — NFC auto-discovery
Parallel	Up to 5 cabinets (DC daisy-chain)

CELLS

Energy module	NVPF sodium-ion (NIB)
Power buffer	EDLC graphene supercapacitor
Raw material	$Na_3V_2(PO_4)_2F_3$ — sodium-ion
Supercap peak	> 50 kW instantaneous

LIFETIME

NIB cycle life	> 8,000 (1C) · 15,000 (0.5C)
Supercap cycles	> 500,000
NIB module warranty	20 yr · 15,000 cycles

CHARGE / DISCHARGE

Max continuous current	280 A (1C)
Method	CC / CV
Communication	RS485 / CANbus
Compatibility	belinus Powerbox S1 & S3

SAFETY

Lithium content	None — fully lithium-free
Thermal runaway	None by design
Chemistry	Inherently safe

WARRANTY

Product	20 yr
Performance	20 yr
Service	20 yr

Notice: specifications are subject to change without notice.